

Medusa Node v1.01: Technical White Paper

Field-Deployable Network Sentinel for Business Continuity

I. Executive Summary

The Medusa Node is a ruggedized, Layer 3 network monitoring unit designed for mission-critical small business environments, specifically "The Daily Grind" café. It provides immediate physical alerting for network latency and gateway cutouts, ensuring POS systems remain operational via manual failover protocols.

II. Hardware Specification

- **Chassis:** Pelican 1050 Micro Case (IP67 Rated for moisture and dust resistance).
- **Compute:** Raspberry Pi SoC optimized for persistent ICMP polling.
- **Thermal Management:** Active RGB Pulse-Width Modulation (PWM) cooling system.
- **Acoustic Alerting:** Integrated GPIO-driven Piezo-acoustic alarm (1kHz frequency).

III. Operational Protocol (The Sentinel Logic)

The unit executes a continuous "Heartbeat" script developed in Python.

1. **Polling:** Performs an ICMP request to the primary gateway every 5 seconds.
2. **Logic:** Verification of the network path at the Network Layer (Layer 3).
3. **Alerting:** Upon three (3) consecutive failed pings, the unit triggers a Layer 1 physical alarm.
4. **Action:** This notifies staff to switch Square/POS systems to "Offline Mode," preventing transaction loss during ISP instability.

IV. Security & Management

- **Administrative Access:** Hardened SSH-2.0 encrypted tunnel.
- **Identity:** Static internal IP mapping within the 192.168.1.0/24 subnet.
- **Traffic Verification:** Real-time packet analysis verifies protocol hierarchy (ARP, IP, TCP/UDP) and ensures non-malicious traffic flow.